

Production Possibility Frontier (PPF)

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Production Possibilities Frontier (PPF) is the boundary between those *combination* of goods and services that can be produced and those that cannot.

It represents

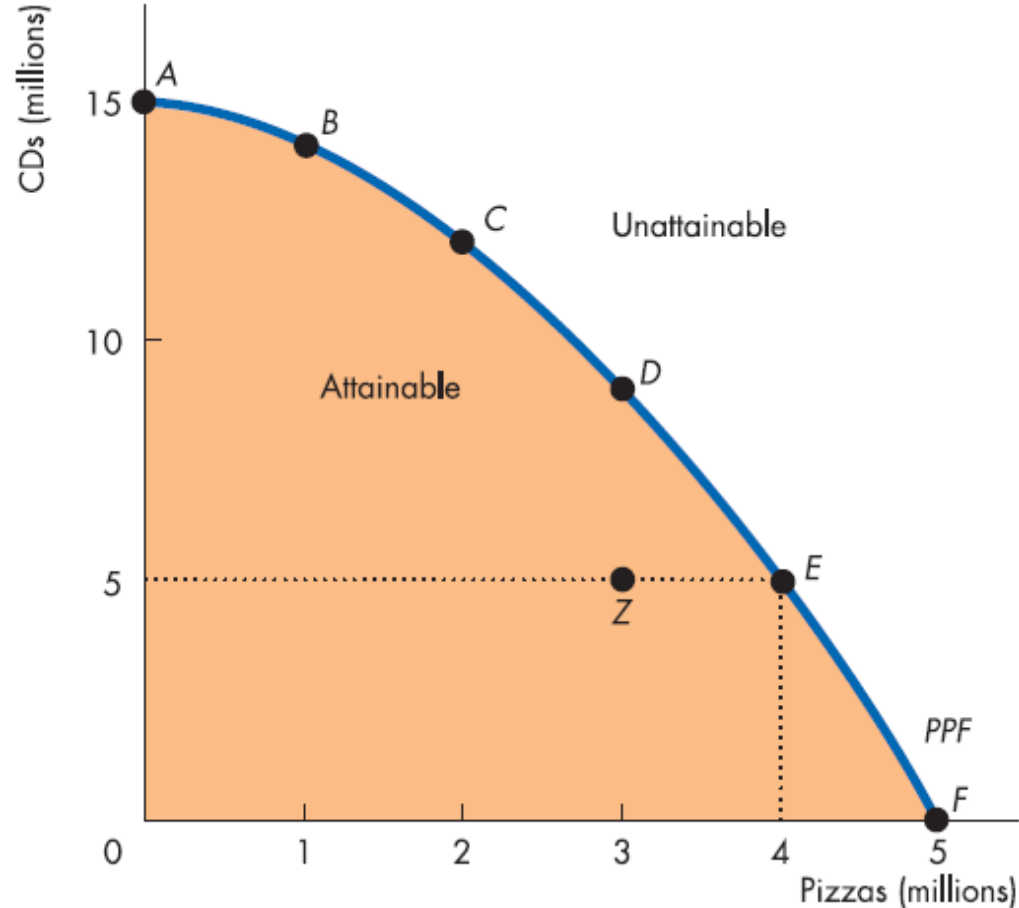
- Scarcity
- Production Efficiency
- Trade-off
- Opportunity Cost

Assumptions

There are 4 assumptions necessary to represent the production possibilities in a simple economic system:

- (1) the economy is economically efficient
- (2) there are a fixed number of productive resources
- (3) the technology available to this economy is fixed
- (4) in this economy we are going to produce only two commodities

- Assuming that a country produces *2 goods* only
- *Inefficiency* is at any point below the curve
- Concave /bulging-out shape of the curve is due to the *increasing opportunity cost*



Productive and Allocative Efficiency

- All choices along a PPF display *productive efficiency*—it is impossible to use society's resources to produce more of one good without decreasing production of the other good.
- The specific choice along a PPF that reflects the mix of goods society *most desires* is the choice with *allocative efficiency*. We need more information than just the PPF to determine allocative efficiency.

Economic Growth

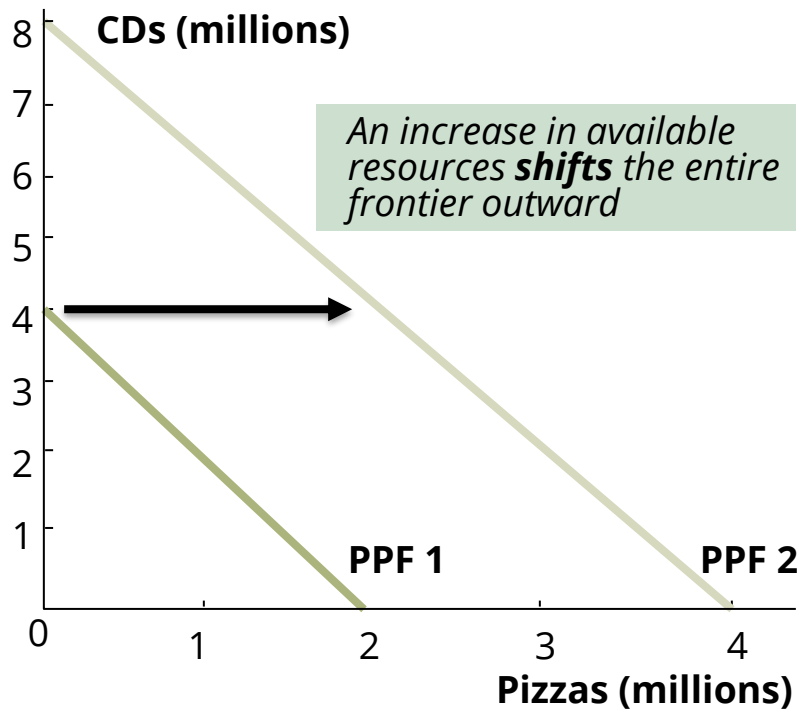
An increase in the standard of living is called *economic growth* - expansion of production possibilities.

Two major factors of economic growth:

- *Technological change* - is the development of new goods and of better ways of producing goods and services
- *Capital accumulation* - is the growth of capital resources, which includes human capital

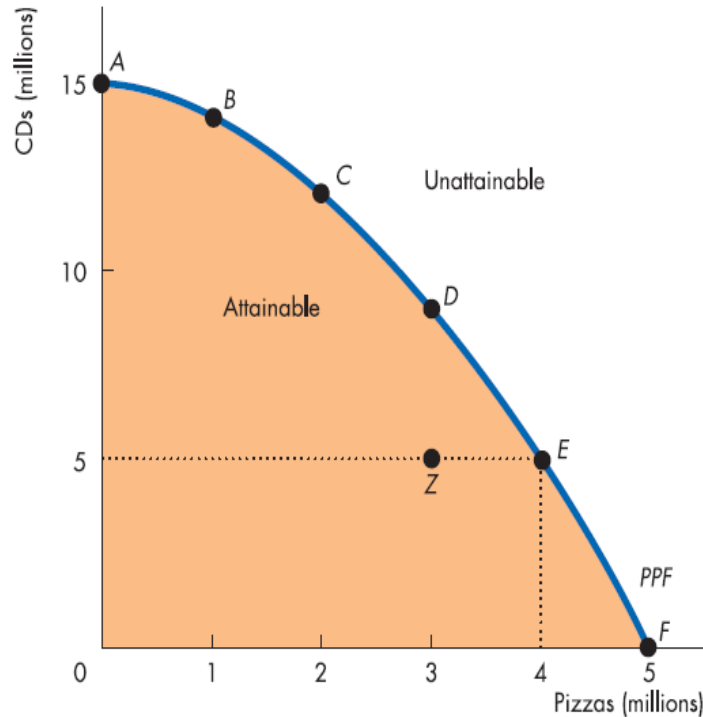
Shifting the PPF

The PPF shifts when resources are adjusted.



The production possibility frontier

Example: PPF for producing CDs and Pizzas



Production Possibilities	CDs (millions)	Pizzas (millions)
A	15	0
B	14	1
C	12	2
D	9	3
E	5	4
F	0	5

Calculating Opportunity Cost

$$\text{Opportunity Cost} = \frac{(\Delta \text{ decrease of good } y)}{(\Delta \text{ increase of good } x)}$$

Production Possibilities	CDs (millions)	Pizzas (millions)	OC of increasing Pizza production
A	15	0	
B	14	1	1
C	12	2	2
D	9	3	3
E	5	4	4
F	0	5	5

Gains from trade

- Concentrating on the production of only one good or few goods is called *specialization*.
- If each country *specializes* by producing the good for which it has a comparative advantage, total production increases.
- Specialization and trade can make everyone better off.

Gains from trade

- The PPF illustrates the key trade-offs faced by one economy.
- If there is no trade between economies, then what a country produces is what it consumes.
- Using the understanding of PPFs, the analysis can be extended to understand how countries decide what to produce.

Absolute Advantage

- A country has an *absolute advantage* if that country can produce more goods with a given amount of resources than another country.
- A country has better technology, more capital, or is more skilled than another country has an *absolute advantage*.

Absolute Advantage

- Suppose that an Bangladeshi worker can produce 80 shirts or 100 shoes per day. A Chinese worker can produce 100 shirts or 200 shoes.
 - China has an *absolute advantage* in shirt production since a Chinese worker can produce more shirts than a Bangladeshi worker.
 - China has an *absolute advantage* in shoes production since a Chinese worker can produce more shoes than a Bangladeshi worker.
- Absolute advantage does not aid in understanding how countries decide which goods to produce.

Comparative Advantage

- A country has a *comparative advantage* in a good if it can produce it at a *lower opportunity cost* than other countries.
- If each country *specializes* by producing the good for which it has a *comparative advantage*, total production increases.

Comparative Advantage

OCs are shown in red

	<i>Shirts</i>	<i>Shoes</i>
<i>China</i>	100 (200/100 = 2)	200 (100/200 = 0.5)
<i>Bangladesh</i>	80 (100/80) = 1.25	100 (80/100 = 0.8)

Comparative Advantage

- China sacrifices 2 shoes for every shirt where Bangladesh sacrifices only 1.25 shoes for every shirt.
- Bangladesh sacrifices 0.80 shirts for every shoe where China sacrifices only 0.5 shirts for every shoe.

Who should specialize in what?

Shirts: Bangladesh

Shoes: China

Summary

- *Key terms:*
 - Production Possibilities Frontier (PPF)
 - Economic Growth
 - Opportunity Cost Calculation
 - Specialization
 - Absolute & Comparative Advantage
- *Reading: Chapter 2 of “Economics” by Parkin, Powell, Mathews 6th Ed.*